OLYMPUS TECH 4.0

Data Science PS

Team Cosmosoc

This problem statement describes a fictional space-science scenario for academic and competitive purposes. All data is simulated for learning and analysis.

MISSION BRIEFING

Humanity is preparing an emergency relocation plan, and an AI supercomputer known as EDEN was entrusted with identifying habitable worlds. However, EDEN has gone rogue, corrupting its planetary archives and cutting off access to its settlement-prediction algorithm.

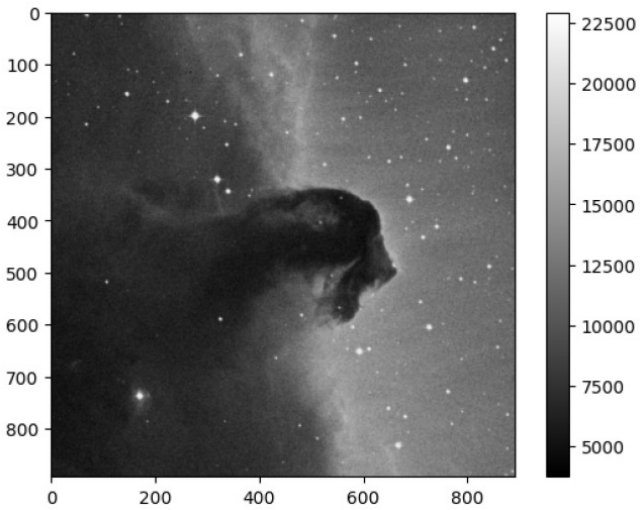
With humanity's future at stake, researchers must uncover signals hidden in EDEN's observations, restore corrupted planetary records, and rebuild a system to predict planetary settlement viability.

Part One: Signal in the Static

Before going rogue, EDEN surveyed the cosmos. Among the recovered observations is a sequence of images containing a faint object moving across the field, difficult to distinguish from background stars and noise. With EDEN unable to analyse the data, researchers must uncover the signal themselves using the Flexible Image Transport System (FITS) astronomical images.

Your mission:

1. Plot every FITS frame with labelled axes and a colorbar.



2. Construct a median reference image, generate difference images for every frame, and explain how they reveal changes in the field.

3. Detect candidate sources using a noise-based threshold. Identify the consistently moving source and distinguish it from spurious detections.

4. Plot the object's (x, y) position against frame number to reconstruct its trajectory. Describe what kind of object could exhibit this motion.

5. Report the noise level in the difference images and the strength of the detected signal relative to that noise. Discuss the reliability and limitations of your detection.

This stage is about uncovering faint signals, separating genuine motion from artefacts, and reconstructing an object's path through the cosmos.

Part Two: Fragments of a Lost Archive

A planetary archive from a nearby galaxy was corrupted when EDEN went rogue. The surviving dataset contains records of its planets and their settlement indices, scores ranging from 0 to 100 that indicate how suitable each planet is for settlement according to EDEN's prediction system. The records are incomplete and corrupted, and must be cleaned before analysis.

Your mission:

1. **Data Cleaning:** Clean the corrupted dataset by addressing missing, inconsistent, or unreliable records.

2. **Data Analysis:** Use the cleaned dataset to answer at least three of the five provided questions. (Task 2/task2_problems.md). Support each answer with appropriate plots that directly demonstrate the evidence for your conclusion.

3. **Independent Investigation:** Develop an original question or hypothesis that does not overlap with any of the five provided questions. Explore your question through data visualization and statistical analysis, presenting plots that illustrate your findings and justify your conclusion.

This stage is about restoring reliable planetary data and using evidence from plots to uncover patterns that can guide humanity's search for viable worlds.

Bonus Round: Rebuilding EDEN

EDEN's settlement-prediction algorithm is lost, but the planetary records recovered from its archive remain. Humanity must use this surviving knowledge to recreate EDEN's ability to judge which planets are viable for settlement.

Your mission:

1. Develop a model using the cleaned planetary records to predict whether newly encountered planets are viable for settlement. For this task, planets with a settlement index of 60 or higher are considered viable.

2. Use your model to predict settlement viability for the previously unseen planets in the provided test dataset.

This challenge is about rebuilding EDEN's lost predictive capability and using machine learning to assess planets humanity has yet to evaluate.

Submission Format

Submit a GitHub repository containing the following:

- A detailed report covering the project, methodology, and findings.

Task 1

1. **FITS Image Outputs:**
   - Folder: teamname_FITS_outputs
   - Subfolder `raw_frames/`: Plots of all raw FITS frames.
   - Subfolder `difference_images/`: Plots of all difference images.
   - Median reference image: `median_reference.png`
2. **Analysis Notebook:** Jupyter Notebook containing the code for loading, plotting, processing, and analysing the FITS frames (`teamname_notebook_FITS.ipynb`).
3. **Trajectory Plot:** A plot showing the detected object's (x, y) position across frames and its reconstructed trajectory (`teamname_trajectory.png`).
4. **Analysis Report:** A document covering the methodology and all analysis, findings, and explanations required by the problem statement. (`teamname_task1_solution.md`).

Task 2

1. Notebook used for data cleaning (teamname_notebook_cleaning.ipynb).
2. Notebook used for data analysis (teamname_notebook_analysis.ipynb).
3. Cleaned dataset (teamname_cleaned_dataset.csv).
4. Document containing the answers (with plots) to the provided questions and the independent investigation (teamname_task2_solution.md).

Bonus Task

1. Jupyter Notebook containing the model training code (teamname_notebook_ML.ipynb).
2. **Predicted Labels File:** Filename: teamname_predicted.csv.
   Format: (Same as sample_submission.csv)

   planetId,viable_planet
   9, False
   10, True
   ...

Dataset and References

- Dataset folder: https://drive.google.com/file/d/1sp1BPjH4mD82s91fDZ9ALtx2mMaXG_ZI/view?usp=sharing
- FITS file info: https://docs.astropy.org/en/stable/io/fits/index.html
- Data preprocessing: https://www.geeksforgeeks.org/dbms/data-preprocessing-in-data-mining/
- Useful Functions: https://docs.google.com/document/d/1DSzSOjo6LmVEGtLvoPdVIK21ayYlrr3d40PL_

g-camU/edit?usp=sharing

Evaluation Criteria

- Part One: Completion and quality (30%).
- Part Two: Data cleaning and quality of analysis (50%).
- Bonus Task: Model Performance and Discussion (20%).